

Finance Engine

User Manual — Every Module, Step by Step,
with a Worked Example You Can Follow

Prepared for GreenLeaf Foods Pvt Ltd (example company used throughout)
Finance Engine v1 · 2026

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1. Introduction — What This Software Does

Finance Engine is a multi-company financial forecasting and controlling platform. It covers the work of three roles that usually sit in separate spreadsheets: a **financial planner** forecasting sales and cash, a **controller** comparing actual spend against budget, and a **management accountant** analyzing product and customer profitability. All three see the same underlying numbers, so a budget approved in one module is the same number a forecast pulls from in another — there is no re-typing figures between tools.

Who this manual is for

You do not need to be an accountant to follow this manual, but it helps to know the basic vocabulary: revenue, expense, budget, variance, contribution margin. Wherever a term first appears, it is explained in plain language before any calculation is shown. If you already know SAP FICO or a similar ERP's Controlling (CO) and Financial Accounting (FI) modules, the concepts will be familiar — Cost Centers, Standard Costing variances, and budget approval chains all follow the same logic you already know, just in a lighter, web-based tool.

How this manual is organized

Every module gets its own chapter with three parts:

1. **What it's for** — the real business question the module answers.
2. **How to use it** — the exact buttons and fields, in order.
3. **How the numbers are actually calculated** — the formula, worked by hand with the example company's real figures, so you can check the software's answer yourself and understand *why* it produced that number, not just trust it blindly.

The same fictional company — **GreenLeaf Foods Pvt Ltd** — is used from Chapter 3 onward, so numbers introduced in one chapter (a sales forecast, a posted actual) are the same numbers referenced in later chapters (cash flow, profitability, financial statements). This is deliberate: in real use, that is exactly how the modules connect to each other.

What Finance Engine deliberately does not do

Being honest about limits up front avoids confusion later:

- There is no double-entry bookkeeping enforcement. Actuals are posted directly against a GL account; the Balance Sheet reports whether assets equal liabilities plus equity rather than assuming it, because nothing forces that to be true.
- There is no live connection to SAP, an ERP, or a bank feed yet. Historical data comes in via manual entry or a bulk Excel/CSV upload.
- Currency conversion is real (see Chapter 14, FX Scenario) but only applied where explicitly stated. Everywhere else, amounts are treated as already being in the company's base currency.
- Bad-debt prediction was deliberately not built, because it needs accounts-receivable ageing (which invoice is how many days overdue) that this system does not track. A feature that faked this from data that cannot support it would give you a confident-looking, meaningless number — worse than no feature at all.

2. Getting Started: Account, Login, Company Setup

2.1 Creating your account

STEPS

1. Open the app. You will land on a Log In / Create Account screen.
2. Click **Create Account**.
3. Enter your name, email, and a password.
4. Click **Create Account**. You are logged in immediately — there is no email verification step.

Your login protects access to the application, but it does not yet separate companies by user — anyone with a valid login can see any company in the system. If you are the only person using this deployment, that is not a problem. If you plan to share this with colleagues, keep that in mind until stricter access control is added.

2.2 Creating your first company

Everything in Finance Engine — every budget, actual, forecast — belongs to a **company**. You can run more than one company (e.g. two subsidiaries) from the same login; a switcher in the top-right corner lets you move between them.

STEPS

1. On first login, you will be prompted to create a company.
2. Enter a **Name** (e.g. "GreenLeaf Foods Pvt Ltd").
3. Enter a **Base Currency** — a three-letter code (USD, EUR, INR, etc.). Every amount you see by default is assumed to already be in this currency.
4. Click **Create**.

2.3 Finding your way around

The left-hand navigation lists every module. They are grouped loosely in the order a finance team would actually use them: forecast revenue and cash first, then control spend against a budget, then analyze what happened, then report it upward. You do not have to use them in that order — the Dashboard on the home page gives you a shortcut view of the most important numbers from several modules at once.

3. The Example Company Used in This Manual

GreenLeaf Foods Pvt Ltd is a fictional packaged-snacks company, base currency USD, used consistently in every chapter below so the numbers connect. If you are following along in your own account, you do not need to recreate this exact company — the point is to understand how the numbers move from one module to the next, using figures small enough to check by hand.

Products

SKU	Product	Unit Price	Unit Variable Cost
CHIP-500	Potato Chips, 500g	\$4.00	\$2.50
COOK-250	Cookies, 250g	\$3.00	\$1.80

Six months of sales history (uploaded via CSV)

Period	Chips Units	Chips Revenue	Cookies Units	Cookies Revenue
2026-01	2,000	\$8,000	1,200	\$3,600
2026-02	2,100	\$8,400	1,150	\$3,450
2026-03	2,300	\$9,200	1,300	\$3,900
2026-04	2,200	\$8,800	1,250	\$3,750
2026-05	2,400	\$9,600	1,400	\$4,200
2026-06	2,600	\$10,400	1,450	\$4,350

Chart of GL accounts

Code	Name	Category	Forecast Role
4000	Sales Revenue	revenue	—
6000	Cost of Goods Sold	expense	—
6100	Marketing Expense	expense	—
6200	Salaries	expense	—
1000	Cash	asset	cash
1100	Accounts Receivable	asset	accounts_receivable
2000	Accounts Payable	liability	accounts_payable
3000	Owner's Equity	equity	—

Approved June budget (Expense type)

GL Account	Budgeted	Actual (posted)
6100 Marketing Expense	\$2,000	\$2,300
6200 Salaries	\$5,000	\$5,000

These figures are referenced again in Chapters 5 through 12 — the marketing overspend in particular is used to show exactly how the variance traffic-light system decides "red" versus "yellow" versus "green."

4. Dashboard

NAVIGATION: DASHBOARD (HOME PAGE)

What it's for

A single screen answering "how is the company doing right now" without opening four different modules. Four KPI tiles, two trend charts, and a live list of anything that looks wrong.

How to use it

STEPS

1. Set the **Fiscal year** field — this scopes Budget Utilization to that year.
2. Set a **Cash runway anchor month** and **Opening balance** if you want the Cash Runway tile populated (it is blank otherwise).
3. Read the four tiles, the two charts below them, and the AI Insights list at the bottom.

How each KPI is calculated

GROSS MARGIN %

```
gross_margin_pct = (sum of contribution margin across products) / (sum of revenue across products) × 100
```

WORKED WITH GREENLEAF'S JUNE NUMBERS

Chips contribution margin = $(\$4.00 - \$2.50) \times 2,600 \text{ units} = \$3,900$

Cookies contribution margin = $(\$3.00 - \$1.80) \times 1,450 \text{ units} = \$1,740$

Total contribution = \$5,640. Total revenue = \$10,400 + \$4,350 = \$14,750.

Gross Margin % = $\$5,640 / \$14,750 \times 100 = 38.2\%$

If a product has no `unit_variable_cost` set, it is excluded from this calculation rather than assumed to have zero cost — an unset cost is not the same fact as a zero cost, and treating it as zero would overstate margin.

BUDGET UTILIZATION %

```
budget_utilization_pct = (total spent across approved budgets) / (total approved budget amount) × 100
```

WORKED EXAMPLE

Approved budget total = \$2,000 + \$5,000 = \$7,000. Actual spent = \$2,300 + \$5,000 = \$7,300.

Budget Utilization % = \$7,300 / \$7,000 × 100 = **104.3%** (over budget)

FORECAST ACCURACY (MAPE)

For each historical month (starting after a short warm-up window), forecast one month ahead using only the data available up to that point, then compare to what actually happened:

```
error_t = |actual_t - forecast_t| / |actual_t|
```

```
MAPE = average(error_t) × 100
```

This is called a **walk-forward backtest**. It never lets the model "see the future" — the forecast for March only ever uses January and February's data, exactly like a real forecast would have to. Lower MAPE means the model's typical error was smaller, so it is a better fit for that specific product's pattern.

CASH RUNWAY (MONTHS)

Starting from the opening balance, add each forecast month's net cash flow (cash in minus cash out) one month at a time. Cash Runway is the number of months until the running balance first goes negative. If it never goes negative within the forecast window, the tile shows "12+ mo" rather than a false precise number.

The two charts

Revenue / Expense / Net Profit Trend plots actual monthly revenue and expense (bars) and net profit (line) for every month you have posted actuals for — however many years that is.

Budget vs. Actual aggregates every GL account's budget and actual for each period into one bar pair, so you can see at a glance whether the company as a whole is over or under budget month to month, before drilling into which account caused it on the Cost Controlling page.

AI Insights

A plain-language list of things worth your attention: budget overruns, unbudgeted spend (an actual posted to an account with no approved budget line at all), and — since Phase 3 — statistical spend/sales anomalies. An anomaly is flagged when the latest month's value is more than 2 standard deviations away from that specific account's own historical average (a **z-score**): $z = \frac{(\text{latest} - \text{mean of prior months})}{\text{population standard deviation of prior months}}$. $|z| \geq 2$ is flagged yellow, $|z| \geq 3$ is flagged red. This is deliberately a statistical rule, not a trained machine learning model — it needs no training data of its own, just at least 4 prior months for that account to compare against.

5. Sales Forecasting

NAVIGATION: SALES FORECAST

What it's for

Projects future revenue for one product at a time, from that product's own sales history. Every other module that needs a revenue number (Cash Flow, Financial Statement Forecast) uses this forecast underneath, summed across all products.

How to use it

STEPS

1. Click **Upload Sales CSV or Excel**. Columns required: `sku, period, quantity, amount, currency` (optional: `product_name, customer_name`). Periods can be `YYYY-MM` or a full date.
2. Pick a **Product** from the dropdown (created automatically from the upload).
3. Pick a **Model** — three statistical models and two machine-learning models, explained below.
4. Set **Periods** (how many months ahead to forecast).
5. Optionally set the product's **Unit variable cost** — this feeds Profitability and Marginal Costing later.

How the statistical models are calculated

MOVING AVERAGE

forecast = average of the last N months (N = window size, default 3), repeated flat for every future month.

WEIGHTED AVERAGE

forecast = weighted average of the last N months, with weights 1, 2, 3 ... N (most recent month weighted heaviest).

EXPONENTIAL SMOOTHING (DEFAULT, $\alpha = 0.3$)

level_1 = first historical value
level_t = $\alpha \times \text{actual}_t + (1 - \alpha) \times \text{level}_{(t-1)}$, for each subsequent month
forecast = the final level, repeated flat for every future month.

WORKED EXAMPLE: CHIPS, EXPONENTIAL SMOOTHING, $\alpha = 0.3$

Starting level = \$8,000 (January)

After February (\$8,400): level = $0.3 \times 8,400 + 0.7 \times 8,000 = \$8,120$

After March (\$9,200): level = $0.3 \times 9,200 + 0.7 \times 8,120 = \$8,344$

After April (\$8,800): level = $0.3 \times 8,800 + 0.7 \times 8,344 = \$8,480.80$

After May (\$9,600): level = $0.3 \times 9,600 + 0.7 \times 8,480.80 = \$8,816.56$

After June (\$10,400): level = $0.3 \times 10,400 + 0.7 \times 8,816.56 = \$9,291.59$

July forecast $\approx$ \$9,292, repeated for every month you ask it to forecast, since exponential smoothing has no concept of a trend beyond the current level.

CONFIDENCE INTERVAL (ALL THREE STATISTICAL MODELS)

residual_std = standard deviation of month-over-month changes in the actual history
margin = $1.96 \times \text{residual_std}$ (a 95% interval)
lower_bound = $\max(\text{forecast} - \text{margin}, 0)$, upper_bound = $\text{forecast} + \text{margin}$

The bound is never allowed to go negative — a forecast revenue below zero is not a meaningful concept.

The two machine-learning models

Random Forest and **Gradient Boosting** are trained on features built from each month's own history: the three preceding months' values (as lags), their rolling average, a plain time index (to capture a trend), and the month itself encoded so December and January read as adjacent rather than eleven months apart. They need at least 6 months of history before they will produce a result — below that, the software refuses to guess rather than return a number trained on almost nothing.

A known, accepted limitation: tree-based models like these can't extrapolate a trend past what they saw in training the way a straight-line projection can. A steadily climbing sales history will get a forecast that plateaus rather than keeps climbing. This is normal for this model family, not a bug — it is exactly why the **Compare models** button exists: run every model's walk-forward MAPE side by side and pick whichever one actually fits your product's pattern best, rather than assuming one model is always right.

Demand Prediction

The **Forecast demand** button runs the exact same models against *units sold* instead of revenue. This matters because price and volume can move independently — a promotion might sell more units at a lower price, which a revenue forecast alone would not distinguish from simply selling fewer units at full price. Demand prediction answers "how many units should we plan to produce," which is a genuinely different question from "how much revenue will we book."

6. Budget Planning

NAVIGATION: BUDGET PLANNING

What it's for

Every company plans spend differently. This module supports four distinct budgeting methods under one roof, plus a formal Manager → Finance → CFO approval chain so a budget cannot be spent against until someone with authority has actually signed off.

6.1 GL Accounts and Cost Centers

Before creating a budget, you need at least one GL account (see Chapter 3's chart of accounts) and, optionally, a Cost Center — a responsibility unit like "Marketing" or "Plant 2" that a line item can be tagged to independently of which GL account it hits. Cost Center tagging is covered fully in Chapter 8.

6.2 The four budget types

Type	What makes it different
Zero-Based	Every line must carry a written justification before the budget can be submitted for approval — spend has to be argued for from zero each cycle, not carried forward by habit.
Flexible	Each line has a fixed portion and a per-unit variable rate. The "budget" it is compared against moves with actual activity — see the worked variance example below.
Rolling	A fixed-size forward window (e.g. 12 months) that rolls forward one month at a time: each roll-forward copies the newest month's line forward and drops the oldest.
Capital	For investments, not period spend — each line captures an investment amount, an annual cash flow, and a useful life, so payback period and ROI can be computed (see below).

6.3 The approval chain

STEPS

1. Create the budget (starts in **draft**).
2. Add line items — while in **draft** or **rejected** status only.
3. Click **Submit for approval** — moves to **pending_manager**. For zero-based budgets, this step is blocked with an error if any line is missing its justification.
4. Each of Manager, Finance, and CFO clicks **Approve** in turn, or **Reject** to send it back to draft for editing and resubmission.
5. After the CFO approves, the budget is **approved** and locked — it now counts toward Budget Utilization and Cost Controlling variance.

Every submission is snapshotted into a version history, so if a budget is rejected, edited, and resubmitted, the earlier line values are still visible rather than silently overwritten.

6.4 Flexible budget variance — worked example

Suppose GreenLeaf's Marketing line in a flexible budget has a fixed portion of \$1,500 plus a variable rate of \$0.20 per unit sold, and June actually sold 4,050 units (chips + cookies combined) at an actual marketing spend of \$2,300.

FLEXIBLE BUDGET FORMULAS

```
flexed_amount = fixed_amount + (variable_rate_per_unit × actual_quantity)
spending_variance = actual_amount - flexed_amount    (efficiency: did we
spend more than the flexed allowance?)
volume_variance = flexed_amount - static_amount    (activity effect: did
more/less volume happen than planned?)
```

WORKED EXAMPLE

```
flexed_amount = $1,500 + ($0.20 × 4,050) = $1,500 + $810 = $2,310
spending_variance = $2,300 - $2,310 = -$10 (favorable — spent $10 less than the flexed allowance)
volume_variance = $2,310 - $1,500 = +$810 (the flexed budget grew because volume was higher
than the static plan assumed)
```

This separates two different questions a single number can't answer on its own: was the overspend because managers spent inefficiently (spending variance), or because the business simply did more volume than planned (volume variance)? In this example, the answer is neither — spending was actually slightly efficient once volume is accounted for.

6.5 Capital budget appraisal — worked example

Suppose GreenLeaf considers a \$50,000 investment in a new packaging line, expected to generate \$15,000 in annual cash flow over a 5-year useful life.

CAPITAL APPRAISAL FORMULAS (SIMPLE, NO DISCOUNTING)

```
payback_period_years = investment / annual_cash_flow
total_cash_flow = annual_cash_flow × useful_life_years
net_gain = total_cash_flow - investment
roi_pct = (net_gain / investment) × 100
average_annual_roi_pct = roi_pct / useful_life_years
```

WORKED EXAMPLE

```
payback_period = $50,000 / $15,000 = 3.33 years
total_cash_flow = $15,000 × 5 = $75,000
net_gain = $75,000 - $50,000 = $25,000
roi_pct = $25,000 / $50,000 × 100 = 50% over the life of the investment
average_annual_roi_pct = 50% / 5 = 10% per year
```

This is deliberately simple appraisal math — no Net Present Value, no discounting for the time value of money. It answers "does this pay for itself, and roughly how well," not "what is this worth today accounting for interest rates." Treat it as a first screen, not a substitute for a proper NPV/IRR analysis on a large investment decision.

7. Cash Flow Forecast

NAVIGATION: CASH FLOW

What it's for

Projects the company's cash balance forward, month by month, combining three sources: forecasted sales (shifted by a collection lag), approved expense budgets, and any manual one-off cash movements you enter directly (payroll runs, loan draws, tax payments).

How to use it

STEPS

1. Set **Start month**, **Periods**, **Collection lag (days)**, and **Opening balance**, then click **Refresh**.
2. To record a one-off item, fill in **Category**, **Direction** (in/out), **Period**, **Amount**, and an optional description, then click **Add**.
3. Every item you have added is listed in a table underneath, with an **Edit** and **Delete** action on each row — click Edit to load a row back into the form above, change any field, and click Save changes; click Delete to remove it entirely. Both actions immediately recompute the forecast below.

How the forecast is calculated

PER PERIOD

$\text{cash_in_total} = \text{cash_in_forecast}$ (from Sales Forecasting, summed across products, shifted forward by the collection lag) + cash_in_manual (your manual "in" items for that month)

$\text{cash_out_total} = \text{cash_out_budget}$ (approved expense budget lines for that month) + cash_out_manual

$\text{net_cash_flow} = \text{cash_in_total} - \text{cash_out_total}$

$\text{closing_balance} = \text{opening_balance} + \text{net_cash_flow}$ (and next month's

$\text{opening balance} = \text{this month's closing}$)

WORKED EXAMPLE, GREENLEAF, JULY, 30-DAY COLLECTION LAG, \$0 OPENING BALANCE

Cash in (forecasted July sales, both products, shifted by 30 days from June's actual sale) $\approx$ \$9,292 (chips, from Chapter 5) + a similarly forecasted cookies figure.

Cash out (July's approved budget) = \$2,000 (marketing) + \$5,000 (salaries) = \$7,000, plus any manual items you add.

$\text{net_cash_flow} = \text{cash_in_total} - \$7,000$.

$\text{closing_balance} = \$0 + \text{net_cash_flow}$, and this closing balance becomes August's opening balance.

Draft or pending-approval budgets are excluded from cash_out_budget on purpose — an unapproved budget is not a real commitment yet, so it should not shrink your projected cash cushion.

8. Cost Controlling & Cost Centers

NAVIGATION: COST CONTROLLING

What it's for

Where actuals get posted against a GL account and compared to what was budgeted, with a traffic-light status so you can scan a long list and immediately see what needs attention.

How to use it

STEPS

1. Under **Post an actual**: pick a GL Account, a Period, an Amount, optionally a Quantity (only used by flexible budgets) and a Cost Center, then click **Post**.
2. Under **Cost Centers**: enter a Code and Name and click **Add cost center** to create one you can tag actuals and budget lines against.
3. The **Budget vs. Actual** table shows every GL account/period combination that has a budget, an actual, or both.
4. The **Cost Center Variance** table shows the same comparison grouped by cost center instead — useful for "how is the Marketing team doing against its budget" rather than "how is this one GL line doing."

How the traffic-light status is decided

VARIANCE %

```
variance_amount = actual_amount - budget_amount  
variance_pct = (variance_amount / budget_amount) × 100
```

DIRECTION-AWARE UNFAVORABLE %

For an **expense** account: $\text{unfavorable_pct} = \text{variance_pct}$ (spending more than budgeted is bad)

For a **revenue** account: $\text{unfavorable_pct} = -\text{variance_pct}$ (earning less than budgeted is bad; earning more is always favorable regardless of size)

$\text{status} = \text{red}$ if $\text{unfavorable_pct} \geq 15\%$, yellow if $\geq 5\%$, otherwise green

WORKED EXAMPLE: GREENLEAF'S JUNE MARKETING OVERSPEND

$\text{variance_amount} = \$2,300 - \$2,000 = \300

$\text{variance_pct} = \$300 / \$2,000 \times 100 = 15\%$

Marketing is an expense account, so $\text{unfavorable_pct} = 15\%$.

$15\% \geq 15\%$, so $\text{status} = \text{red}$.

Note that Salaries, posted exactly on budget (\$5,000 vs \$5,000), gets $\text{variance_pct} = 0\%$ and $\text{status} = \text{green}$. A favorable variance of any size is always green — the thresholds only bite in the unfavorable direction, since underspending or overachieving is never something you need a warning about.

Budget Consumption

CONSUMPTION %

$\text{consumption_pct} = (\text{spent} / \text{budget_amount}) \times 100$

$\text{status} = \text{red}$ if $\geq 100\%$, yellow if $\geq 80\%$, otherwise green

Unlike Budget vs. Actual, this tracks any budget regardless of approval status — a department manager watching their spend against a working budget during the approval cycle does not want to wait for CFO sign-off to see how close they are to the limit.

Cost Center Accounting

A cost center is independent of GL account category — the same "Marketing" cost center could carry both a marketing-expense line and, in principle, a revenue line, though in practice cost centers are almost always used for expense tracking. Tagging is optional everywhere (posting an actual, adding a budget line, or a bulk statement upload with a `cost_center_code` column) — only lines actually tagged appear in the Cost Center Variance report; untagged spend is not

silently bucketed into a fabricated "Unassigned" total, since that would misrepresent data that was never actually attributed anywhere.

9. Profitability & Marginal Costing (CVP)

NAVIGATION: PROFITABILITY

What it's for

Answers "which products and customers are actually profitable," and, at the company level, the classic cost-volume-profit (CVP) questions: what is our break-even revenue, how much cushion do we have above it, and how much does a change in revenue move our profit.

Contribution margin per product

FORMULAS

```
contribution_per_unit = unit_price - unit_variable_cost
contribution_margin_total = contribution_per_unit × quantity_sold
contribution_margin_pct = contribution_per_unit / unit_price × 100
```

WORKED EXAMPLE: CHIPS, JUNE

```
contribution_per_unit = $4.00 - $2.50 = $1.50
contribution_margin_total = $1.50 × 2,600 = $3,900
contribution_margin_pct = $1.50 / $4.00 × 100 = 37.5%
```

If a product's `unit_variable_cost` has never been set, its contribution fields are shown as null, not zero — an unset cost is a data gap, and treating it as zero would make an uncosted product look artificially more profitable than it might be. The Marginal Costing summary lists any such product's SKU under `uncosted_product_skus` so it is not silently dropped either.

Marginal Costing / CVP analysis

FORMULAS

```
contribution_margin_ratio = total contribution margin / total revenue  
(company-wide, weighted across all costed products)  
break_even_revenue = total fixed costs / contribution_margin_ratio  
margin_of_safety = actual_revenue - break_even_revenue  
margin_of_safety_pct = margin_of_safety / actual_revenue × 100  
degree_of_operating_leverage = total_contribution_margin /  
net_operating_income
```

WORKED EXAMPLE, GREENLEAF, JUNE, ASSUMING \$4,000 TOTAL FIXED COSTS (RENT + FIXED SALARIES)

Total revenue = \$10,400 + \$4,350 = \$14,750. Total contribution = \$3,900 (chips) + \$1,740 (cookies) = \$5,640.

contribution_margin_ratio = \$5,640 / \$14,750 = 0.3824 (38.24%)

break_even_revenue = \$4,000 / 0.3824 = **\$10,460**

margin_of_safety = \$14,750 - \$10,460 = \$4,290

margin_of_safety_pct = \$4,290 / \$14,750 × 100 = **29.1%**

net_operating_income = \$5,640 - \$4,000 = \$1,640

degree_of_operating_leverage = \$5,640 / \$1,640 = **3.44**

The operating leverage figure means: a 1% change in revenue would move net operating income by roughly 3.44% in the same direction, at this specific revenue level. That sensitivity is highest near the break-even point and falls as a company sells further above it — which is exactly what the margin of safety percentage is telling you at the same time.

Customer Churn Risk

FORMULA

```
risk_ratio = months_since_last_order / customer's own  
average_order_interval_months  
risk_level: high if ratio ≥ 2.0, medium if ≥ 1.3, otherwise low
```

This compares a customer to *their own* normal buying rhythm, not a fixed rule like "no order in 3 months." A customer who orders twice a year and a customer who orders twice a month need completely different definitions of "overdue" — this ratio gives each customer their own yardstick.

It deliberately is not a trained classifier: this system has no record of subscription cancellations or "churn" events to train one on, and inventing a label to train against would not be real machine learning, just a confident-looking guess.

10. Standard Costing (Variance Analysis)

NAVIGATION: STANDARD COSTING

What it's for

The classic manufacturing variance method: compare what a unit of production was supposed to cost (the standard) against what it actually cost, broken into eight distinct variances so you know exactly which input — material price, material usage, labor rate, labor efficiency, or overhead — drove any difference.

How to use it

STEPS

1. Enter the **standard cost sheet** for a product once: material standard price and quantity, labor standard rate and hours, variable and fixed overhead rates, and budgeted fixed overhead.
2. Post **production actuals** each period: units produced, and the actual material price/quantity, labor rate/hours, and overhead incurred.
3. The variance table computes all eight variances automatically for every product with both a standard and an actual on file.

The eight variances, worked with an example

Suppose GreenLeaf's Chips standard cost sheet says: material standard price \$0.80/kg, standard 0.3kg per unit; labor standard rate \$15/hour, standard 0.05 hours per unit. In June, 2,600 units were produced using actual material at \$0.85/kg and 0.31kg per unit, and actual labor at \$14.50/hour for 0.052 hours per unit.

MATERIAL VARIANCES

$\text{material_price_variance} = (\text{standard_price} - \text{actual_price}) \times \text{actual_quantity_used}$
 $\text{material_quantity_variance} = (\text{standard_quantity_allowed} - \text{actual_quantity_used}) \times \text{standard_price}$
where $\text{standard_quantity_allowed} = \text{standard_qty_per_unit} \times \text{units_produced}$

WORKED

$\text{actual_quantity_used} = 0.31 \times 2,600 = 806 \text{ kg}$
 $\text{material_price_variance} = (\$0.80 - \$0.85) \times 806 = -\40.30 (unfavorable — paid more per kg than standard)
 $\text{standard_quantity_allowed} = 0.3 \times 2,600 = 780 \text{ kg}$
 $\text{material_quantity_variance} = (780 - 806) \times \$0.80 = -\$20.80$ (unfavorable — used more material than standard allows)

LABOR VARIANCES

$\text{labor_rate_variance} = (\text{standard_rate} - \text{actual_rate}) \times \text{actual_hours_worked}$
 $\text{labor_efficiency_variance} = (\text{standard_hours_allowed} - \text{actual_hours_worked}) \times \text{standard_rate}$

WORKED

$\text{actual_hours_worked} = 0.052 \times 2,600 = 135.2 \text{ hours}$
 $\text{labor_rate_variance} = (\$15 - \$14.50) \times 135.2 = +\67.60 (favorable — paid less per hour than standard)
 $\text{standard_hours_allowed} = 0.05 \times 2,600 = 130 \text{ hours}$
 $\text{labor_efficiency_variance} = (130 - 135.2) \times \$15 = -\$78.00$ (unfavorable — took more labor hours than standard allows)

OVERHEAD VARIANCES (SAME LOGIC, APPLIED TO VARIABLE AND FIXED OVERHEAD SEPARATELY)

$\text{variable_overhead_spending_variance} = (\text{standard_variable_oh_rate} - \text{actual_rate}) \times \text{actual_hours}$

$\text{variable_overhead_efficiency_variance} = (\text{standard_hours_allowed} - \text{actual_hours}) \times \text{standard_variable_oh_rate}$

$\text{fixed_overhead_budget_variance} = \text{budgeted_fixed_overhead} - \text{actual_fixed_overhead}$

$\text{fixed_overhead_volume_variance} = (\text{standard_hours_allowed} - \text{budgeted_hours}) \times \text{standard_fixed_oh_rate}$

Every variance follows one sign convention: **positive is favorable, negative is unfavorable**. Adding all eight together for a product gives its total cost variance for the period — the single number a controller usually leads with, before drilling into which of the eight components actually caused it.

11. Financial Statements

NAVIGATION: FINANCIAL STATEMENTS

What it's for

Builds an Income Statement and Balance Sheet directly from the actuals you have posted — no separate data entry. It also supports bulk-uploading historical statements (however many years you have) and downloading every report as Excel, PDF, or PowerPoint.

Uploading historical statements

STEPS

1. Click **Upload historical statements**.
2. Provide an Excel or CSV file with columns: `gl_account_code`, `category`, `period`, `amount` (optional: `gl_account_name`, `currency`, `cost_center_code`).
3. GL accounts and cost centers referenced by code that do not already exist are created automatically. There is no limit on how many years of history a single upload can cover — three years, four years, or more, all in one file.

How the Income Statement is built

FORMULA

`total_revenue` = sum of every actual posted to a revenue-category GL account within the selected period range

`total_expense` = sum of every actual posted to an expense-category GL account within the same range

`net_profit` = `total_revenue` - `total_expense`

WORKED EXAMPLE, GREENLEAF, JUNE ONLY

Revenue (4000 Sales) = \$10,400 + \$4,350 = \$14,750

Expense (6100 Marketing + 6200 Salaries, ignoring COGS for simplicity here) = \$2,300 + \$5,000 = \$7,300

net_profit = \$14,750 - \$7,300 = **\$7,450**

How the Balance Sheet is built

FORMULA

For each of Assets, Liabilities, and Equity: the balance of an account as of a given date is the running sum of every actual ever posted to it, up to and including that date — not just the current period.

`is_balanced = |total_assets - (total_liabilities + total_equity)| ≤ $0.01`

Because there is no double-entry enforcement anywhere in this system, `is_balanced` and `difference` report the truth rather than assume it — if actuals were not posted consistently on both sides of a transaction, the report will show exactly how far off it is rather than silently forcing a balance. This is a deliberate design choice: a report that always claims to balance regardless of the underlying data would be actively misleading.

12. Financial Statement Forecasting

NAVIGATION: FINANCIAL STATEMENTS → FINANCIAL STATEMENT FORECAST SECTION

What it's for

Projects the Income Statement and Balance Sheet forward, using two selectable methods that answer slightly different questions.

Method 1: Driver-based (the default)

INCOME STATEMENT DRIVERS

`revenue_forecast(month)` = sum of the Sales Forecasting projection (Chapter 5) across every product, for that month

`expense_forecast(month)` = sum of approved budget lines on expense-category GL accounts, for that month

BALANCE SHEET DRIVERS

`Accounts Receivable` = `revenue_forecast` / 30 × `DSO_days` (Days Sales Outstanding)

`Accounts Payable` = `expense_forecast` / 30 × `DPO_days` (Days Payable Outstanding)

`Cash` = reuses the Cash Flow Forecast's closing balance (Chapter 7), seeded from the actual balance of accounts tagged `forecast_role = cash`

`Equity` = actual equity as of the day before the forecast starts, plus the running total of forecasted net income (assumes no dividends or new capital – the honest default, since neither is modeled anywhere in this system)

Every other account (not tagged cash / AR / AP) is carried forward flat at its last actual balance.

WORKED EXAMPLE, GREENLEAF, JULY, DSO 45 DAYS, DPO 30 DAYS

Using the July revenue forecast built up across products in Chapter 5 (chips $\approx$ \$9,292, plus a similarly forecast cookies figure) and the July approved expense budget (\$7,000 from Chapter 3):

Accounts Receivable $\approx$ revenue_forecast / 30 $\times$ 45 (one and a half months of revenue sitting uncollected)

Accounts Payable = \$7,000 / 30 $\times$ 30 = \$7,000 (exactly one month of expense, since DPO = 30 days = 1 month)

Method 2: Historical Trend

The alternative for when you have bulk-uploaded multiple years of historical statements (Chapter 11) and want to project forward from that trend directly, the way you would by hand in a spreadsheet — pull three or four years of statements and extrapolate — rather than from the sales-forecast-plus-budget drivers above.

FORMULA

revenue_forecast and expense_forecast are each computed independently by applying the same forecasting models from Chapter 5 (moving average, weighted average, exponential smoothing, or one of the two ML models) directly to that category's own full monthly actuals history — however many years exist, with no cap.

Pick one of the two ML models here specifically if you want the projection to pick up an actual multi-year growth or decline trend — the three statistical models will always project a flat number, exactly as in Chapter 5. This method only produces an Income Statement; the Balance Sheet still uses the driver-based method's DSO/DPO logic, since a trend-based balance sheet (which deals in running balances, not period-over-period deltas) is a different kind of problem that was not built rather than shipped half worked-out.

13. Scenario Planning

NAVIGATION: SCENARIO PLANNING

What it's for

A named "what if" — what if sales grow 10% faster than the base forecast, or expenses come in 5% lower — computed side by side against the exact same base-case Financial Statement Forecast from Chapter 12, so the comparison is real and reproducible rather than two disconnected guesses.

How to use it

STEPS

1. Enter a **Name**, a **Sales growth %**, and an **Expense growth %** (use a negative number for a decrease), then click **Create Scenario**.
2. Select the scenario to see its Income Statement and Balance Sheet forecast next to the base case (0% growth on both).

FORMULA

`sales_factor = 1 + sales_growth_pct / 100`

`expense_factor = 1 + expense_growth_pct / 100`

Every month's `revenue_forecast` and `expense_forecast` (Chapter 12) is multiplied by these factors. The Balance Sheet's Cash line is recomputed with the same factors applied to the Cash Flow Forecast's inputs, so Cash does not silently stay at the base case while everything else moves.

WORKED EXAMPLE: "OPTIMISTIC" SCENARIO, +10% SALES, -5% EXPENSE, APPLIED TO GREENLEAF'S JULY DRIVER-BASED FORECAST

`sales_factor = 1.10`, `expense_factor = 0.95`

Base `revenue_forecast` ≈ \$9,292 (chips) → scenario revenue ≈ \$9,292 × 1.10 = **\$10,221**

Base `expense_forecast` = \$7,000 → scenario expense = \$7,000 × 0.95 = **\$6,650**

Base net profit ≈ \$2,292 → scenario net profit ≈ \$10,221 - \$6,650 = **\$3,571**

14. FX Scenario

NAVIGATION: FX SCENARIO

What it's for

The currency-risk counterpart to Scenario Planning — instead of asking "what if the business grows," it asks "what if the exchange rate moves." It converts your non-base-currency sales into your base currency using the most recent rate on file, then shows what the same foreign-currency revenue would be worth if that rate moved by a hypothetical percentage.

How to use it

STEPS

1. Under **Exchange Rates**, enter a From currency, To currency, Date, and Rate, then click **Save rate**. Saving the same currency pair and date again updates the rate in place rather than creating a duplicate.
2. Set a date range and a **Shock %** (negative for a depreciation, positive for an appreciation), then click **Run scenario**.

FORMULA

For each non-base-currency sale in the range: the system finds the most recent exchange rate on or before that sale's period — never a future-dated rate, and never an invented 1:1 fallback if none exists.

`base_amount = native_amount × rate_used`

`shocked_amount = native_amount × rate_used × (1 + shock_pct / 100)`

`impact = sum(shocked_amount) - sum(base_amount)`

WORKED EXAMPLE

Suppose GreenLeaf also sells €1,000 worth of product in Europe in a given month, and a EUR→USD rate of 1.10 is on file dated on or before that month.

$\text{base_amount} = €1,000 \times 1.10 = \$1,100$

With a -10% shock (the euro weakens against the dollar):

$\text{shocked_amount} = €1,000 \times 1.10 \times 0.90 = \990

$\text{impact} = \$990 - \$1,100 = -\$110$ (the company would book \$110 less in dollar terms for the exact same €1,000 of sales)

Any currency with no rate on file at all is listed separately and excluded from the totals — it is not assumed to convert at 1:1, since that would misrepresent an unknown as a fact. This tool is deliberately scoped to sales actuals only; it does not retroactively convert every other report in the system (Financial Statements, KPIs, Cash Flow) to multi-currency, since that would be a much larger and riskier change than a single scenario tool.

15. Monte Carlo Simulation

NAVIGATION: SALES FORECAST → MONTE CARLO SIMULATION SECTION

What it's for

The point forecast and confidence interval in Chapter 5 use one fixed-width band for every future month, no matter how far out you look. In reality, the further into the future you forecast, the more can go differently — a projection six months out should carry a wider range of plausible outcomes than a projection one month out. Monte Carlo simulation captures that by running many randomized trials instead of one formula.

How it works

FORMULA

1. Compute the point forecast for each future month exactly as in Chapter 5.
2. Compute `residual_std`, the same historical volatility measure used for the confidence interval.
3. Run 1,000 trials. In each trial, draw a random shock for every month from a normal distribution centered on zero with that standard deviation, and add each month's shock to the running total of all previous shocks in that trial (a "random walk") — this is what makes uncertainty compound the further out you go, rather than stay constant.
4. Add each trial's accumulated shock to that month's point forecast, and floor the result at zero (revenue cannot be negative).
5. Across all 1,000 trial values for a given month, report the 10th percentile (`p10`), 50th percentile / median (`p50`), and 90th percentile (`p90`).

The practical read: `p10` to `p90` is your realistic range of outcomes for that month, and you will see that range get wider the further into the future the month is — exactly the behavior a single confidence interval cannot represent.

16. AI Insights (Recommendation Engine)

NAVIGATION: DASHBOARD → AI INSIGHTS SECTION

What it's for

A running list of plain-language flags pulled automatically from the other modules, so you do not have to remember to go check each one separately.

What it checks, and how

Insight	Rule
Budget overrun / shortfall	The same variance and threshold logic as Chapter 8 (5% yellow / 15% red, direction-aware by GL category).
Unbudgeted spend	An actual is posted to a GL account/period with no approved budget line at all for that combination.
Budget-consumption warning	The same 80% yellow / 100% red thresholds as Chapter 8's Budget Consumption.
Forecasted sales decline	The Sales Forecasting model (Chapter 5) projects a lower figure than recent actuals.
Statistical spend/sales anomaly	The z-score method described in Chapter 4 — flags what is unusual for that <i>specific</i> account or product's own history, not a company-wide fixed threshold.

This is explicitly a rule-based engine, not a trained machine-learning model or a language model. Every flag traces back to one of the formulas already explained earlier in this manual — there is no hidden logic behind it.

17. Reports & Exports

Excel exports

Every major report — Income Statement, Balance Sheet, Financial Statement Forecast, and Sales Forecast — has a **Download Excel** button producing a real `.xlsx` file with the same figures shown on screen, organized into sheets.

PDF and PowerPoint Board Reports

From the Financial Statements page, the **Board Report** section produces a formatted document — not a raw table dump — combining the Dashboard's four KPIs (Chapter 4), the Income Statement, and the Balance Sheet for whatever date range you have set. **Download PDF** gives a print-ready document; **Download PowerPoint** gives the same content as slides, useful for dropping straight into a board deck.

Bulk data upload

Two upload paths exist: a sales CSV/Excel upload (Chapter 5, for revenue history) and a historical statements Excel/CSV upload (Chapter 11, for the full chart of accounts across however many years of history you have). Both find-or-create the underlying records (products, GL accounts, cost centers) by their code, so re-uploading a file with some already-known codes and some new ones only creates what is actually new.

18. Audit Trail

NAVIGATION: AUDIT TRAIL

What it's for

A record of who changed what, and when — every budget action through its full approval lifecycle, every actual posted, every GL account and cost center created or edited, every scenario created or deleted, and every manual cash item added, edited, or deleted.

How to use it

STEPS

1. Optionally filter by **Entity type** (Budget, Actual, GL Account, Cost Center, Scenario, Cash Item).
2. Read the table: When, Entity, Action, Actor (the logged-in user who performed it), and a plain-language Summary of what happened.

Each entry is a short, human-readable sentence rather than a raw before/after data dump — "Approved budget 'Marketing Budget' as finance (Finance Fay)" is meant to be read directly, not decoded.

19. Quick Reference: Every Formula in One Place

For when you just need the formula, not the walkthrough. See the referenced chapter for a fully worked numeric example of each.

Metric	Formula	Chapter
Gross Margin %	$\Sigma \text{contribution margin} / \Sigma \text{revenue} \times 100$	4, 9
Budget Utilization %	$\Sigma \text{spent} / \Sigma \text{approved budget} \times 100$	4
Forecast Accuracy (MAPE)	$\text{avg}(\text{actual} - \text{forecast} / \text{actual}) \times 100$, walk-forward	4
Contribution margin / unit	$\text{unit_price} - \text{unit_variable_cost}$	9
Contribution margin %	$\text{contribution} / \text{unit_price} \times 100$	9
Break-even revenue	$\text{fixed_costs} / \text{contribution_margin_ratio}$	9
Margin of safety %	$(\text{revenue} - \text{break_even}) / \text{revenue} \times 100$	9
Operating leverage	$\text{contribution_margin} / \text{net_operating_income}$	9
Variance %	$(\text{actual} - \text{budget}) / \text{budget} \times 100$	8
Flexible: spending variance	$\text{actual} - \text{flexed_amount}$	6
Flexible: volume variance	$\text{flexed_amount} - \text{static_amount}$	6
Capital: payback period	$\text{investment} / \text{annual_cash_flow}$	6
Capital: ROI %	$\text{net_gain} / \text{investment} \times 100$	6
Material price variance	$(\text{std_price} - \text{actual_price}) \times \text{actual_qty}$	10
Material quantity variance	$(\text{std_qty_allowed} - \text{actual_qty}) \times \text{std_price}$	10
Labor rate variance	$(\text{std_rate} - \text{actual_rate}) \times \text{actual_hours}$	10
Labor efficiency variance	$(\text{std_hours_allowed} - \text{actual_hours}) \times \text{std_rate}$	10
Exponential smoothing	$\text{level} = \alpha \cdot \text{actual} + (1 - \alpha) \cdot \text{level}$	5
Confidence interval	$\text{forecast} \pm 1.96 \times \text{residual_std}$	5

Anomaly z-score	$(\text{latest} - \text{mean}) / \text{population_std}$, flagged at $ z \geq 2$ / ≥ 3	4
Cash flow: closing balance	$\text{opening} + \text{cash_in_total} - \text{cash_out_total}$	7
Days Sales Outstanding (AR)	$\text{revenue_forecast} / 30 \times \text{DSO_days}$	12
Days Payable Outstanding (AP)	$\text{expense_forecast} / 30 \times \text{DPO_days}$	12
Scenario growth factor	$1 + \text{growth_pct} / 100$, applied multiplicatively	13
FX conversion	$\text{native_amount} \times \text{latest_rate_on_or_before}(\text{period})$	14
Churn risk ratio	$\text{months_since_last_order} / \text{avg_order_interval}$	9
Monte Carlo band	$\text{point_forecast} + \text{cumulative random shock}$, percentiled across 1,000 trials	15

Finance Engine User Manual — generated 2026. This manual describes the software as built; if a feature's behavior in your deployment differs from what is written here, trust the software's own on-screen explanatory text first, since features do occasionally evolve after a manual is written.